

Supply Chain Logistics Manager - Non-Functional Requirements

Confidential - Internal Use Only

Version 1.0

1. Performance Requirements

NFR-LOG-001: Shipment dashboard page load < 2 seconds with 1,000+ active shipments displayed.

NFR-LOG-002: Carrier rate shopping API must return all carrier quotes within 3 seconds (parallel fan-out).

NFR-LOG-003: Live tracking map must render 5,000 simultaneous shipment markers without frame drops.

NFR-LOG-004: Warehouse barcode scan-to-database-update latency < 200ms to support rapid picking workflows.

NFR-LOG-005: Route optimization for 50-stop route must complete within 10 seconds.

2. Scalability & Capacity

NFR-LOG-006: System must handle 2 million shipments per month with 100,000 new shipments per day during peak season (Q4).

NFR-LOG-007: Tracking event ingestion: 500,000 events per hour from carrier webhooks and IoT devices.

NFR-LOG-008: Warehouse system must support 8 warehouses with 500 concurrent scanner sessions total.

NFR-LOG-009: Report generation for 12-month data range must complete within 30 seconds.

3. Security & Compliance

NFR-LOG-010: SOC 2 Type II compliance for all cloud infrastructure. Annual audit required.

NFR-LOG-011: Carrier API credentials encrypted at rest using AWS Secrets Manager with automatic rotation every 90 days.

NFR-LOG-012: Denied party screening must complete within 5 seconds and must use lists updated within last 24 hours.

NFR-LOG-013: C-TPAT (Customs-Trade Partnership Against Terrorism) tier 2 compliance for US imports.

NFR-LOG-014: All customer shipment data isolated in multi-tenant architecture. Cross-tenant data access technically impossible.

4. Availability & Disaster Recovery

NFR-LOG-015: System uptime SLA: 99.9% (max 43.8 minutes downtime per month). Planned maintenance during Sunday 2-4am UTC.

NFR-LOG-016: Multi-AZ AWS deployment with auto-scaling. RTO < 15 minutes, RPO < 1 minute.

NFR-LOG-017: Carrier API failover: if primary carrier API unreachable for 30 seconds, switch to backup endpoint/method.

NFR-LOG-018: Offline capability for warehouse scanners: local cache with sync when connectivity restored within 4 hours.

5. Integration Reliability

NFR-LOG-019: Carrier webhook delivery: at-least-once processing with idempotency keys.

Duplicate event detection within 24-hour window.

NFR-LOG-020: Dead letter queue for failed carrier API calls. Automatic retry: 3 attempts with exponential backoff (1s, 5s, 25s).

NFR-LOG-021: ERP integration health check every 5 minutes. Alert if sync lag exceeds 15

minutes.